

IMPACT OF BREAST CANCER AWARENESS CAMPAIGNS ON SCREENING PARTICIPATION AND CANCER DETECTION: A THREE-YEAR EXPERIENCE AT HOSPITAL AL-SULTAN ABDULLAH (HASA)

Nurul'Ain Abdul Thani, Nazimah Ab Mumin, Marlina Tanty Ramli Hamid, Roqiah Fatmawati Abdul Kadir
Department of Radiology, Faculty of Medicine, Universiti Teknologi Mara, Sungai Buloh, Selangor, Malaysia



BACKGROUND

- Breast cancer screening participation remains suboptimal in many Asian populations despite year-round availability of mammography.
- Breast Cancer Awareness Month (BCAM) campaigns may increase screening uptake; however, their impact on the risk profile of attendees and cancer detection is less well characterised.
- This study compared screening participation and cancer detection during BCAM and non-BCAM periods at a Malaysian tertiary centre.

OBJECTIVES

Primary: To assess the diagnostic performance of breast cancer screening at HASA for three years.

Secondary:

- To compare breast cancer detection rates during BCAM and non-BCAM screenings.
- To assess participation of high-risk women during BCAM and non-BCAM screenings.

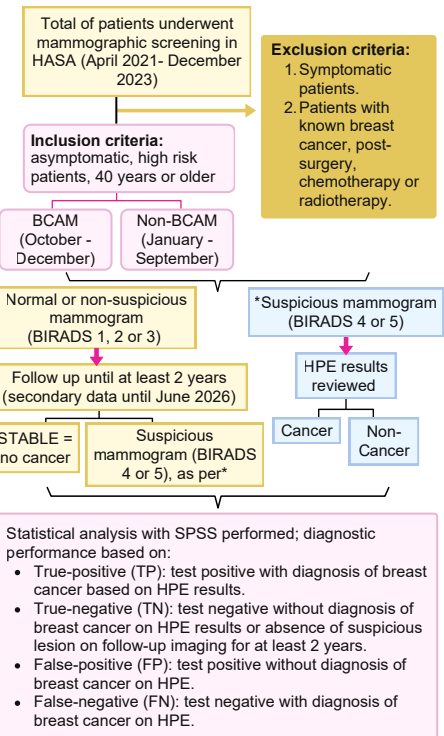
METHODS

Design: Retrospective cohort diagnostic performance study

Setting: Radiology Department, HASA

Study period: Secondary data included from 2021 until June 2026

Participants: Asymptomatic patients who attended mammographic screenings during the specified time frame.



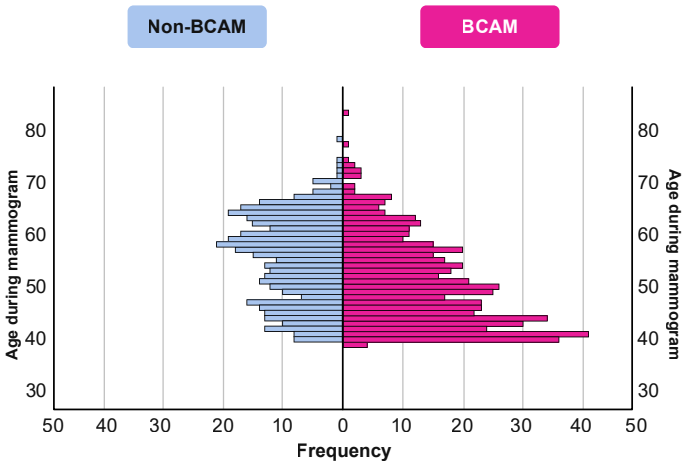
Reference standard and statistical analysis:

- Screening examinations were categorised according to **ACR BI-RADS**.
- Participants were grouped into **BCAM** (Oct-Dec) and **non-BCAM** (Jan-Sep) periods.
- Histopathology was the **reference standard** for biopsied lesions; nonbiopsied cases were followed for up to 2 years where available.
- Screening participation, risk profile, biopsy utilisation and cancer detection rates were **compared between the two periods**.

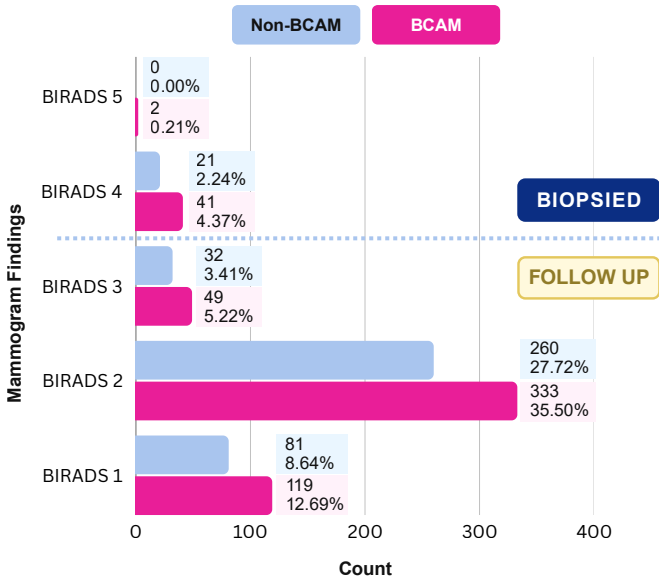
RESULTS

Total Participation During BCAM vs Non-BCAM

	Race					Total
	Malay	Chinese	Indian	Others (Malaysian)	Foreigner	
Non-BCAM	359	17	12	2	5	395
BCAM	508	17	17	4	1	547
Total	867	34	29	6	6	942



Most Common BIRADS Category During Mammogram Screening



BIOPSIED

HPE RESULTS

BCAM: 3 cancers (CDR = 5.48 per 1,000)

- 2 high-risk women

NON-BCAM: 2 cancers (CDR = 5.06 per 1,000)

- Both average-risk women

COMPARISON

No significant difference between periods (p=1.00)

FOLLOW UP

MAMMOGRAM FOLLOW-UP

238 (Non-BCAM) + 308 (BCAM) = 546 out of 942 did not return

INTERVAL CANCERS

4 detected

- 2 high-risk women

OVERALL CDR | 5.31 PER 1,000

$$CDR = \frac{\text{Number of cancers detected}}{\text{Number of screening exams}} \times 1000$$

- More high-risk women attended during non-BCAM than during BCAM (30.6% vs 23.2%; p-value 0.02).
- Sensitivity, specificity, positive predictive value and negative predictive value for screening mammogram were 0.56, 0.94, 0.08 and 0.99.

CONCLUSION / TAKE-HOME MESSAGE

1. MAIN FINDING

- Awareness campaigns increased screening activity but **did not significantly increase cancer detection rate** per screening.
- Women screened outside the campaign period were **more often high-risk**, suggesting BCAM attracts women who less likely to go screening.

2. CLINICAL MEANING

Higher screening activity does not always translate to higher detection rate.

Campaigns can engage **both high-risk and average-risk** women.

3. NEXT STEP

Sustain year-round screening strategies with awareness campaigns to **broaden participation**.

REFERENCES

- Bhandari, A. K. C., Htay, Z. W., Parvin, R., Murakami, M., & Abe, S. K. (2024). Prevalence of Breast Cancer Screening in Asia: Systematic Review and Meta-analysis. In Asian Pacific Journal of Cancer Prevention (Vol. 25, Number 10, pp. 3379–3391). Asian Pacific Organization for Cancer Prevention. <https://doi.org/10.31557/APJCP.2024.25.10.3379>
- Chotai, N., Renganathan, R., Uematsu, T., Wang, J., Zhu, Q., Rahmat, K., Pradaranon, V., Fong, J. C. Y., Choridah, L., & Chang, J. M. (2025). Breast Cancer Screening in Asian Countries: Epidemiology, Screening Practices, Outcomes, Challenges, and Future Directions. In Korean Journal of Radiology (Vol. 26, Number 8, pp. 743–758). Korean Radiological Society. <https://doi.org/10.3348/kjr.2025.0338>
- Htay, M. N. N., Su, T. T., & Donnelly, M. (2024). Risk-Stratified Breast Cancer Screening in Malaysia: Challenges and Opportunities. In Asian Pacific Journal of Cancer Prevention (Vol. 25, Number 3, pp. 785–791). Asian Pacific Organization for Cancer Prevention. <https://doi.org/10.31557/APJCP.2024.25.3.785>
- Said, Z. M., Sutan, R., Malaysia, K., Yaakob, J., Bandar, L., & Razak, T. (2021). Are We Doing Enough in Promoting Breast Cancer Screening: An Evaluation of Malaysia Breast Cancer Screening Program. In Malaysian Journal of Public Health Medicine (Vol. 21, Number 3).

ACKNOWLEDGEMENTS

The study was approved by the Faculty Ethics Review Committee of the Faculty of Medicine, University Teknologi MARA and Department of Research, Innovation & Industrial Linkages (PJI), Hospital Al-Sultan Abdullah, UiTM, in 2025 [600-HUiTM (PJI. 5/2/2480)].

Special appreciation to Mdm Nurzatul Azmira Arbian and the Mammogram Team / Radiographers.

Study concepts and design came from AP Dr. Nazimah AM and Dr. Roqiah Fatmawati AK.

Literature research and data collection were done by Nurul 'Ain AT.

The data and statistical analysis were performed by AP Dr. Nazimah AM, AP Dr. Marlina Tanty RH, and Nurul 'Ain AT.

No funding or conflict of interest from the authors.

